

Reinforcement Learning

Off-policy Monte Carlo Prediction/Control

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Monte Carlo with ϵ -soft policies

On-policy first-visit MC control (for ϵ -soft policies), estimates $\pi \approx \pi_*$

Algorithm parameter: small $\epsilon > 0$

Initialize:

$\pi \leftarrow$ an arbitrary ϵ -soft policy

$Q(s, a) \in \mathbb{R}$ (arbitrarily), for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$

$Returns(s, a) \leftarrow$ empty list, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$

Repeat forever (for each episode):

Generate an episode following π : $S_0, A_0, R_1, \dots, S_{T-1}, A_{T-1}, R_T$

$G \leftarrow 0$

Loop for each step of episode, $t = T-1, T-2, \dots, 0$:

$G \leftarrow \gamma G + R_{t+1}$

Unless the pair S_t, A_t appears in $S_0, A_0, S_1, A_1, \dots, S_{t-1}, A_{t-1}$:

Append G to $Returns(S_t, A_t)$

$Q(S_t, A_t) \leftarrow \text{average}(Returns(S_t, A_t))$

$A^* \leftarrow \arg\max_a Q(S_t, a)$ (with ties broken arbitrarily)

For all $a \in \mathcal{A}(S_t)$:

$$\pi(a|S_t) \leftarrow \begin{cases} 1 - \epsilon + \epsilon/|\mathcal{A}(S_t)| & \text{if } a = A^* \\ \epsilon/|\mathcal{A}(S_t)| & \text{if } a \neq A^* \end{cases}$$

Policy Improvement Guarantees with ϵ -soft Policy

- $\pi'(a|s)$ is ϵ -greedy policy
 - ▶ (Greedy) chooses $A = \arg \max_a q_\pi(s, a)$ with probability $1 - \epsilon + \frac{\epsilon}{|\mathcal{A}(s)|}$
 - ▶ (Exploration) choose $A \neq \arg \max_a q_\pi(s, a)$ with probability $\frac{\epsilon}{|\mathcal{A}(s)|}$

$$q_\pi(s, \pi'(s)) = \sum_a \pi'(a|s) q_\pi(s, a)$$

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Policy Improvement Guarantees with ϵ -soft Policy (cont'd)

- So we have $q_\pi(s, \pi'(s)) \geq v_\pi(s)$ for all $s \in \mathcal{S}$.

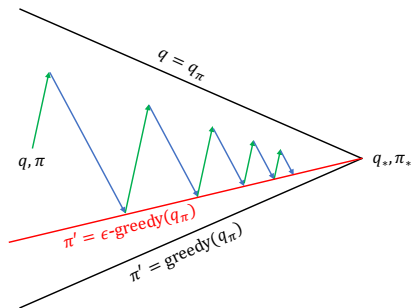
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- So we have $q_\pi(s, \pi'(s)) \geq v_\pi(s)$ for all $s \in \mathcal{S}$.
- By the policy improvement theorem, π' is better than, or as good as, π (see Pg 78 in Sutton & Barto).
- But, both π and π' are ϵ -soft policies... Does it make sense?

On-policy Monte Carlo Control with ϵ -soft Policy



- For any ϵ -soft policy π , any ϵ -greedy policy with respect to q_π is guaranteed to be better than or equal to π .

On-policy vs. Off-policy

- **On-policy**: evaluate or improve a policy that is used to make decisions
 - ▶ Data generating policy & policy to improve are the same
- **Off-policy**: evaluate or improve a target policy from data generated by a different behavior policy
 - ▶ Target policy $\pi(a|s)$: a policy being learned and optimized
 - ▶ Behavior policy $b(a|s)$: policy that interacts with the environment
i.e., $\{s_1, a_1, r_2, s_2, a_2, \dots, s_T\} \sim b$

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- On-policy is a special case of off-policy! Why? e.g., $\pi(a|s) = b(a|s)$

Off-policy Learning

“Learning one policy while following another policy”

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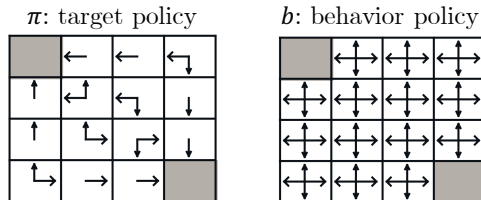
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- Why important?
 - ▶ Learn from observing behaviors of humans or other agents
 - ▶ Re-use data generated from old policies
 - ▶ Learn about multiple policies while following one policy
e.g., optimal policy while following exploratory policy, counterfactual, etc.

Off-policy Prediction

- Given data from behavior policy b , we want to learn the value function v_π of target policy π .

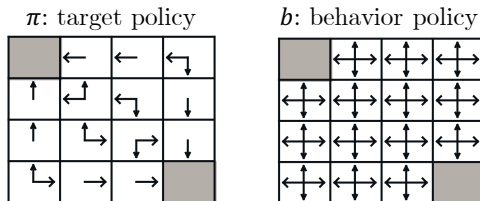
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- (Coverage)** For any $a \in \mathcal{A}(s)$, $\pi(a|s) > 0$, we require $b(a|s) > 0$.



Off-policy Prediction

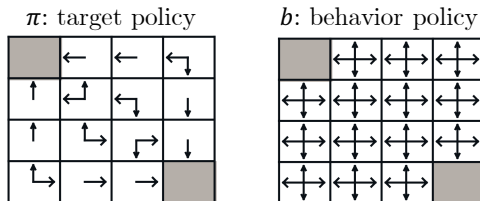
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- How do you learn v_π from following b ?

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- How do you learn v_π from following b ? Importance sampling!

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Importance Sampling

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But, in off-policy setting, you can't...
- **Idea of importance sampling**: draw a sample from a behavior policy and re-weight using **importance weights** so that the correct distribution (policy) is targeted

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- Want to estimate $\mathbb{E}_\pi[X]$

Importance Sampling Derivation

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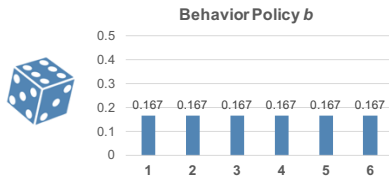
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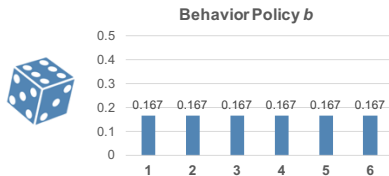
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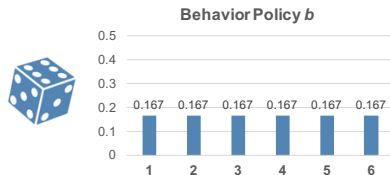
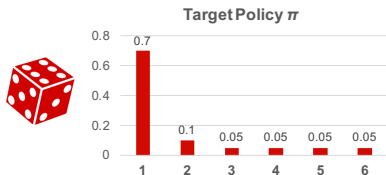
Suppose you roll 3 times

- $X_1 = 1$. $\pi(1) = 0.7, b(1) = 0.167$



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Suppose you roll 3 times

- $X_1 = 1$. $\pi(1) = 0.7, b(1) = 0.167$
- $X_2 = 2$. $\pi(2) = 0.1, b(2) = 0.167$
- $X_3 = 1$. $\pi(1) = 0.7, b(1) = 0.167$

$$\mathbb{E}_{\pi}[X] \approx \frac{1}{n} \sum_{i=1}^n \rho(x_i) x_i = \frac{(1 \times \frac{0.7}{0.167}) + (2 \times \frac{0.1}{0.167}) + (1 \times \frac{0.7}{0.167})}{3} = 3.2$$

Off-policy Prediction using Importance Sampling

- Likelihood of trajectory under policy π

$$\begin{aligned} & \Pr\{A_t, S_{t+1}, A_{t+1}, \dots, S_T \mid S_t, A_{t:T-1} \sim \pi\} \\ &= \pi(A_t | S_t) p(S_{t+1} | S_t, A_t) \pi(A_{t+1} | S_{t+1}) \cdots p(S_T | S_{T-1}, A_{T-1}) \\ &= \prod_{k=t}^{T-1} \pi(A_k | S_k) p(S_{k+1} | S_k, A_k), \end{aligned}$$

Off-policy Prediction using Importance Sampling

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- Compute importance weights

$$\rho_{t:T-1} \doteq \frac{\prod_{k=t}^{T-1} \pi(A_k | S_k) p(S_{k+1} | S_k, A_k)}{\prod_{k=t}^{T-1} b(A_k | S_k) p(S_{k+1} | S_k, A_k)} = \prod_{k=t}^{T-1} \frac{\pi(A_k | S_k)}{b(A_k | S_k)}.$$

Off-policy Prediction using Importance Sampling

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- Value function using “ordinary” and “weighted” importance sampling

$$V(s) \doteq \frac{\sum_{t \in \mathcal{T}(s)} \rho_{t:T(t)-1} G_t}{|\mathcal{T}(s)|}, \quad V(s) \doteq \frac{\sum_{t \in \mathcal{T}(s)} \rho_{t:T(t)-1} G_t}{\sum_{t \in \mathcal{T}(s)} \rho_{t:T(t)-1}}.$$

Off-policy Monte Carlo Prediction

Off-policy MC prediction (policy evaluation) for estimating $Q \approx q_\pi$

Input: an arbitrary target policy π

Initialize, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$:

$Q(s, a) \in \mathbb{R}$ (arbitrarily)

$C(s, a) \leftarrow 0$

Loop forever (for each episode):

$b \leftarrow$ any policy with coverage of π

Generate an episode following b : $S_0, A_0, R_1, \dots, S_{T-1}, A_{T-1}, R_T$

$G \leftarrow 0$

$W \leftarrow 1$

Loop for each step of episode, $t = T-1, T-2, \dots, 0$, while $W \neq 0$:

$G \leftarrow \gamma G + R_{t+1}$

$C(S_t, A_t) \leftarrow C(S_t, A_t) + W$

$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \frac{W}{C(S_t, A_t)} [G - Q(S_t, A_t)]$

$W \leftarrow W \frac{\pi(A_t|S_t)}{b(A_t|S_t)}$

Off-policy Monte Carlo Control

Off-policy MC control, for estimating $\pi \approx \pi_*$

Initialize, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$:

$Q(s, a) \in \mathbb{R}$ (arbitrarily)

$C(s, a) \leftarrow 0$

$\pi(s) \leftarrow \operatorname{argmax}_a Q(s, a)$ (with ties broken consistently)

Loop forever (for each episode):

$b \leftarrow$ any soft policy

Generate an episode using b : $S_0, A_0, R_1, \dots, S_{T-1}, A_{T-1}, R_T$

$G \leftarrow 0$

$W \leftarrow 1$

Loop for each step of episode, $t = T-1, T-2, \dots, 0$:

$G \leftarrow \gamma G + R_{t+1}$

$C(S_t, A_t) \leftarrow C(S_t, A_t) + W$

$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \frac{W}{C(S_t, A_t)} [G - Q(S_t, A_t)]$

$\pi(S_t) \leftarrow \operatorname{argmax}_a Q(S_t, a)$ (with ties broken consistently)

If $A_t \neq \pi(S_t)$ then exit inner Loop (proceed to next episode)

$W \leftarrow W \frac{1}{b(A_t|S_t)}$

Next class: TD Learning
(Chapter 6 of Sutton & Barto)